AMRITA VISHWA VIDYAPEETHAM
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MOFChat: Explainable, Data-Grounded MOF Recommendation with Validated LLM Planning, Multi-Objective Optimisation and Deep-Learning Surrogates

22AIE452 – Data Driven Material Modeling and Simulation · Final Review

Team_09

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Introduction

Metal–Organic Frameworks (MOFs) are crystalline porous materials made by linking metal ions with organic molecules, creating microscopic cages. Their tunable pores make them candidates for **carbon capture, hydrogen and methane storage, gas separation and catalysis**.

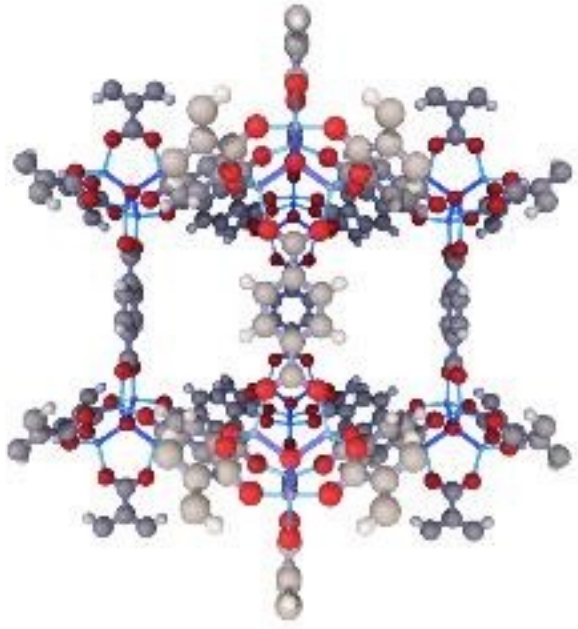
Selecting the right MOF is hard: the CoRE-MOF database alone has **12,020 MOFs**, and a good choice trades off several properties (surface area, pore size, density, metal sites) and needs domain knowledge.

Problem statement

Existing LLM agents (ChatMOF) make MOF search conversational, but they write code that can fail, depend on paid models, hit token limits and return numbers that nobody checks against the data.

Our aim

A conversational assistant that answers MOF questions exactly from the full database, ranks trade-offs, predicts properties with uncertainty and explanation, runs on free or offline models, and verifies its own answers.



Example MOF framework: metal nodes (cages) joined by organic linkers

12,020
MOFs

65
metals

27 %
zero accessible area

Base Paper: ChatMOF (Kang & Kim, Nat. Commun., 2024)

What the base paper did

- GPT-4 agent that plans tool calls (ReAct)
- Search: LLM writes pandas code over the CoRE-MOF table
- Prediction: MOFTransformer (one fine-tuned model per property)
- Generation: genetic algorithm with PORMAKE

96.9 % search accuracy

95.7 % prediction accuracy

87.5 % generation accuracy

https://www.nature.com/articles/s41467-024-48998-4

Drawbacks reported in the base paper

L1 No intrinsic materials knowledge (LLM only plans tools)

L2 Token-context limits: 4 % of search tasks failed

L3 Reasoning and logic errors in generated code

L4 Restricted search space (≈100 parent structures)

L5 Narrow topology coverage (9 topologies)

L6 One pretrained model needed per property

L7 General, non-fine-tuned LLM (GPT-4)

L8 High compute cost (GA + simulations)

L9 Generated / returned results are not validated

Objectives and Our Novelty

Objectives

1. Exact answers over all 12,020 MOFs with free or local LLMs

2. Safe LLM planning: validation, self-correction, fallback

3. Multi-objective ranking with NSGA-II

4. Deep-learning and ML surrogates with uncertainty and SHAP

5. Knowledge grounding by retrieval (RAG)

6. Measured results: benchmark, held-out set, classification and regression metrics

Base-paper limitation	MOFChat novelty
No materials knowledge	RAG over curated notes + ChatMOF paper
Token limits	SQL filters first; LLM sees ≤ 10 verified rows
Logic errors in code	LLM writes a JSON plan; validator checks and repairs it
Small search space	All 12,020 MOFs searched and ranked (NSGA-II)
Only 9 topologies	Every CoRE-MOF structure, 65 metals
One model per property	Own XGBoost / FT-Transformer surrogates
Proprietary GPT-4	14 free or local LLMs with automatic fallback
High computation	Millisecond surrogates instead of simulations
No validation	Answer validator + conformal intervals + SHAP + benchmark

Literature Review (2024–2026) — 1 / 3

#	Paper	Authors	Year / Venue	Method	Contribution	Limitation	Use in MOFChat
1	**ChatMOF: an AI system for predicting and generating MOFs using LLMs**	Y. Kang, J. Kim	2024, Nat. Commun. 15, 4705	GPT-4 agent writes pandas code; MOFTransformer; genetic algorithm	96.9 / 95.7 / 87.5 % on search / prediction / generation	Token limits, code errors, answers not checked	Base paper; we use validated JSON plans + SQL
2	**Augmenting LLMs with chemistry tools (ChemCrow)**	A. M. Bran, S. Cox, O. Schilter, C. Baldassari, A. D. White, P. Schwaller	2024, Nat. Mach. Intell. 6	GPT-4 + 18 expert tools in a reason-act loop	Planned real syntheses; found a new chromophore	Needs GPT-4; tool outputs not cross-checked	Tool-calling idea; our tools are deterministic and tested
3	**MOFDiff: coarse-grained diffusion for MOF design**	X. Fu, T. Xie, A. S. Rosen, T. Jaakkola, J. Smith	2024, ICLR	Equivariant diffusion over building blocks + assembly	Valid, novel MOFs for CO2 capture	Needs simulation to validate candidates	Generation is future work for MOFChat
4	**TableRAG: million-token table understanding with LMs**	S.-A. Chen, L. Miculicich, J. M. Eisenschlos et al.	2024, NeurIPS	Schema and cell retrieval before the LLM	Scales table QA to huge tables	Numeric filters still judged by the LLM	We filter with SQL first; LLM sees ≤ 10 rows
5	**Chain-of-Table: evolving tables in the reasoning chain**	Z. Wang, H. Zhang, C.-L. Li et al.	2024, ICLR	LLM plans table operations step by step	State of the art on WikiTQ, TabFact	No domain validation of operations	We validate every planned operation
6	**Detecting hallucinations in LLMs using semantic entropy**	S. Farquhar, J. Kossen, L. Kuhn, Y. Gal	2024, Nature 630, 625–630	Entropy over meanings of sampled answers	Detects confabulations without task data	Misses consistent errors; many samples	We verify answers against data deterministically

Literature Review (2024–2026) — 2 / 3

#	Paper	Authors	Year / Venue	Method	Contribution	Limitation	Use in MOFChat
7	**Harnessing LLMs to collect and analyze MOF property data (L2M3)**	W. Lee, Y. Kang, T. Bae, J. Kim	2025, JACS 147, 3943–3958	LLM text mining of > 40,000 articles	Experimental MOF property database + synthesis recommender	Extraction errors from text	Source of real labels for future uptake surrogates
8	**MOF-ChemUnity: literature-informed LLMs for MOF research**	T. M. Pruyn, A. Aswad, S. T. Khan, J. Huang, R. Black, S. M. Moosavi	2025, JACS	Multi-agent LLM links literature names to CSD structures	Literature-grounded answers, knowledge graph	Large infrastructure	Our RAG grounds answers in notes + ChatMOF paper
9	**Accurate predictions on small data with a tabular foundation model (TabPFN)**	N. Hollmann, S. Müller, L. Purucker et al., F. Hutter	2025, Nature 637, 319–326	Transformer pretrained on synthetic tables; in-context prediction	Beats tuned GBDT on ≤ 10k-sample tables	Limited to ~10k rows / 500 features	Candidate surrogate for our 8k-row tables
10	**TabM: advancing tabular DL with parameter-efficient ensembling**	Y. Gorishniy, A. Kotelnikov, A. Babenko	2025, ICLR	MLP + BatchEnsemble	Beats attention-based tabular DL	Generic benchmarks	Supports our finding: FT-Transformer ≈ trees
11	**LLaMP: LLM made powerful for materials knowledge retrieval**	Y. Chiang, E. Hsieh, C.-H. Chou, J. Riebesell	2025, EMNLP (arXiv 2024)	Hierarchical ReAct agents over Materials Project	Grounded answers + confidence metric	Needs Materials Project API	Same grounding idea for MOFs with a local DB
12	**System of agentic AI for the discovery of MOFs (MOFGen)**	T. J. Inizan, S. Yang, …, K. A. Persson, O. M. Yaghi	2025, arXiv 2504.14110	LLM proposes compositions; diffusion + quantum-chemistry agents	Synthesizable MOF candidates	Heavy compute; preprint	Agent + verification trend; we verify against the DB

Literature Review (2024–2026) — 3 / 3

#	Paper	Authors	Year / Venue	Method	Contribution	Limitation	Use in MOFChat
13	**L²M³OF: a large language multimodal model for MOFs**	J. Cui, F. Wu, H. Zhao et al.	2025, arXiv 2510.20976	Crystal encoder projected into an LLM	Beats text-only LLMs on property prediction	Needs structure files + fine-tuning	We stay tabular; multimodal is future work
14	**Conversational LLM agent for synthesis of MOF catalysts (MOFsyn Agent)**	J. Lin et al.	2025, ACS Nano 19, 23840–23858	RAG + ML agent guiding synthesis	Faster Ni@UiO-66(Ce) catalyst optimisation	Single reaction system	RAG + ML in a MOF agent, like MOFChat
15	**The rise of generative AI for MOF design and synthesis**	C. Duan, A. Nandy, …, S. M. Moosavi, Z. Zheng	2025, arXiv 2508.13197 (perspective)	Review of VAEs, diffusion, LLM agents	Roadmap to closed-loop discovery	Feasibility and data gaps	Places MOFChat at the screening stage
16	**LitMOF: LLM multi-agent for literature-validated MOF database correction**	H. Kim, D. Kim, J. Kim	2025, arXiv 2512.01693 (rev. 2026)	Agents check CSD structures against papers	189,567 structures; 9,227 entries repaired	Preprint	Data quality matters: we audit CoRE-MOF in ETL
17	**Enhancing spatial reasoning in LLMs for MOF structure prediction (MOF-LLM)**	M. Pan, J. Li, P. Liu et al.	2026, arXiv 2601.09285	Qwen-3 8B with spatial pretraining, SFT and RL	35.78 % match rate, 0.04 s per structure	Low match rate	Small open LLMs can be specialised (cf. our Llama 3 path)
18	**An LLM agent for end-to-end computational materials discovery (MAESTRO)**	Y. Chen, J. Huang, Y. Liu et al.	2026, arXiv 2608.20434	Agent mines literature, builds DB, screens MOFs	Finds wet-flue-gas separation candidates	Preprint	End-to-end agents; MOFChat is the verified query layer

Research Gap from the Literature

Gap	Evidence in 2024–2026 work	What MOFChat does
Answers are rarely verified against data	ChatMOF, ChemCrow, LLaMP trust the LLM's final text; semantic entropy needs many samples.	Deterministic answer validator: IDs, numbers and comparison claims checked against the database.
Dependence on proprietary LLMs	Most agents use GPT-4.	14 free / local models with fallback; works fully offline (Llama 3).
Few honest benchmarks for planning	Accuracy is reported on the authors' own question sets.	64-question dev set + 20 held-out phrasings + model comparison + ablation.
Tabular DL vs trees on MOF descriptors	TabPFN / TabM / Grinsztajn compare on generic data.	FT-Transformer vs XGBoost vs GP vs MLP on MOF regression and classification.

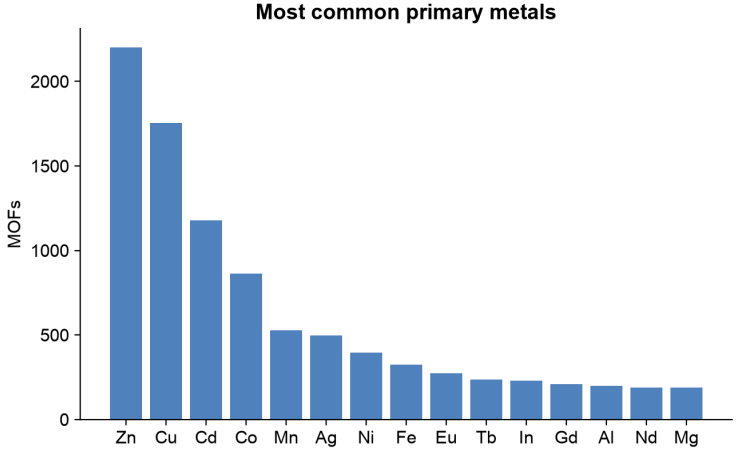
Dataset: CoRE MOF 2019 and Preprocessing

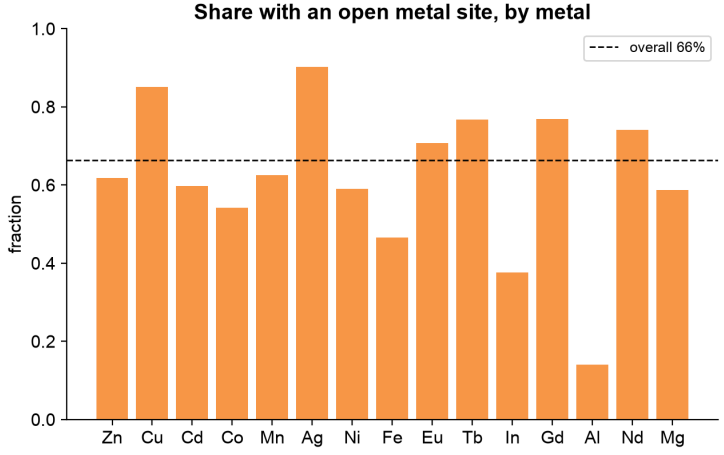
Category	Attributes
Identification	MOF_ID (CSD refcode)
Pore geometry	LCD, PLD, LFPD (Å)
Density	Density (g/cm³)
Surface area	Accessible (m²/g, m²/cm³), non-accessible
Porosity	Void fraction, pore volume, non-accessible volume
Chemistry	Metal type, open metal site, OMS type

12,020 experimental MOFs × 17 columns, computed with Zeo++ and a 3.31 Å (N_2) probe.

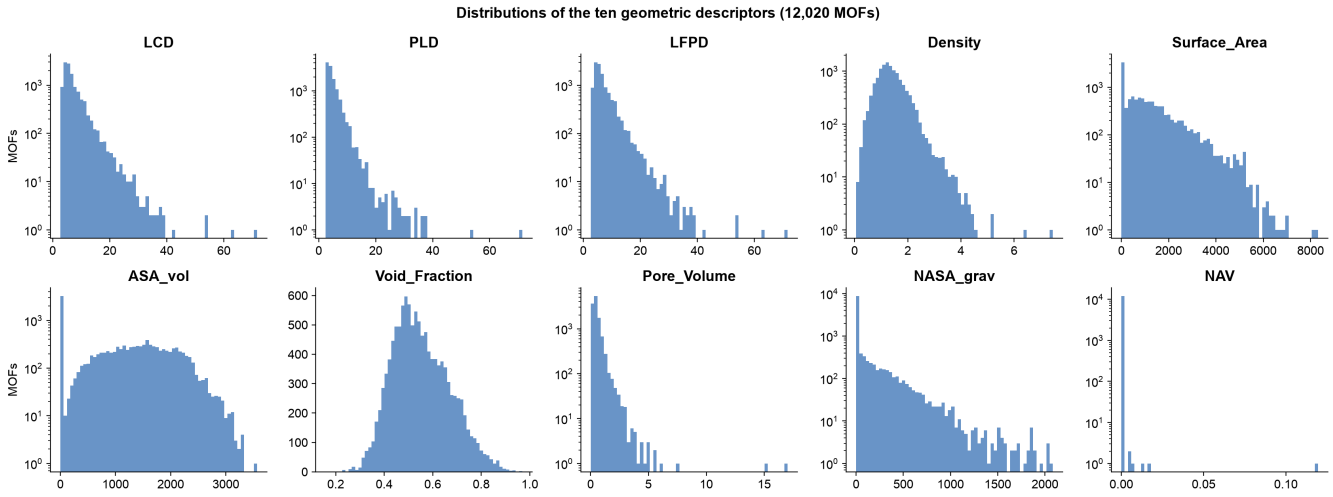
Preprocessing (ETL) findings

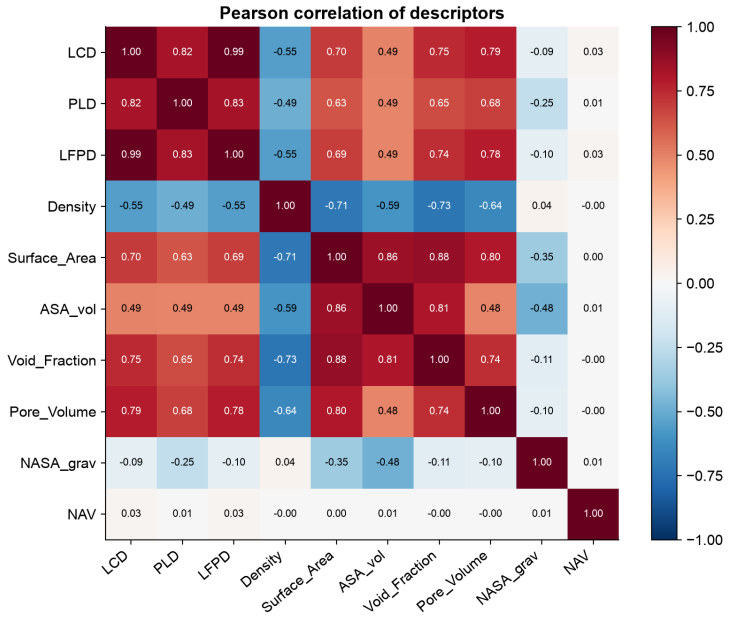
Check	Finding → action
Missing values / duplicates	none
Two column pairs identical	dropped duplicates
Pore volume = void fraction / density	derived; documented
Zero surface area (3,252)	all have PLD < 3.31 Å → physics rule
Density header said cm³/g	values are g/cm³ → label fixed
2,087 multi-metal MOFs	separate mof_metals table
Feature scaling	log1p + z-score (Preprocessor)





Feature Extraction (1): Descriptor Distributions and Correlations



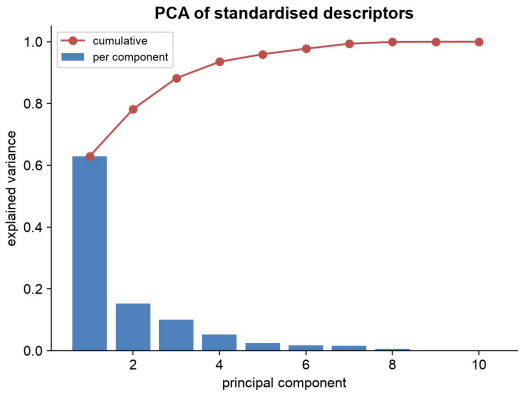


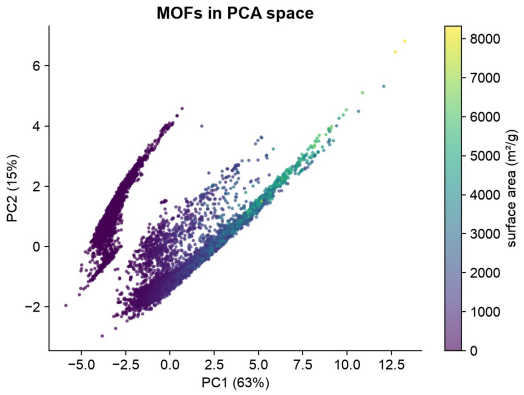
Pearson correlation of the ten descriptors

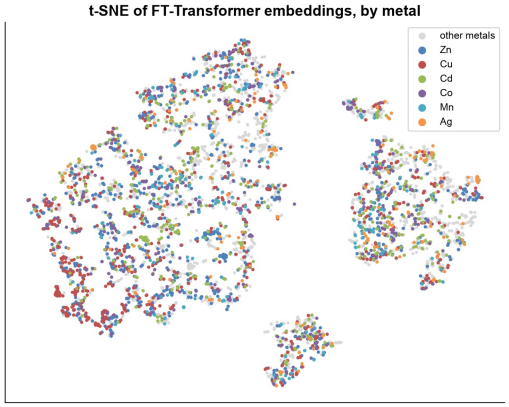
What the graphs show

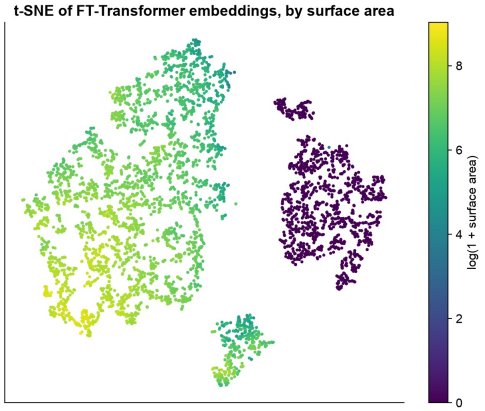
- Most descriptors are right-skewed (log scale) → log1p + z-score normalisation before the FT-Transformer
- 27 % of MOFs have exactly zero accessible surface area (the spike at 0)
- Strongest correlations: LCD–LFPD: +0.99; Surface_Area–Void_Fraction: +0.88; Surface_Area–ASA_vol: +0.86; PLD–LFPD: +0.83
- Pore volume is derived (void fraction / density), so it is excluded when predicting void fraction (no leakage)

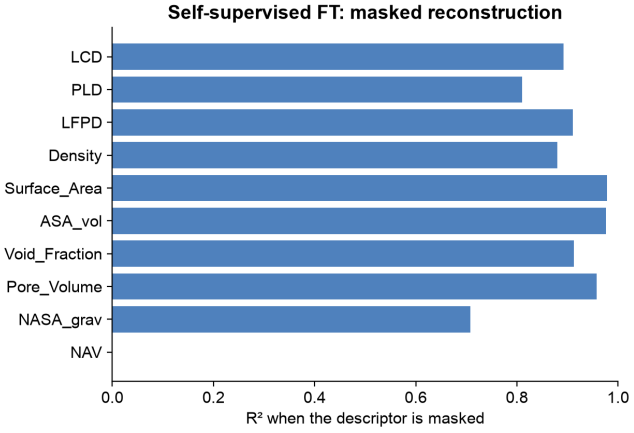
Feature Extraction (2): PCA, Learned Embeddings, Reconstruction

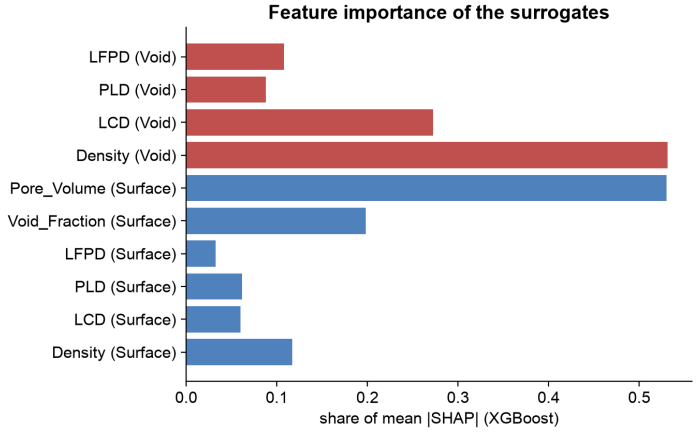






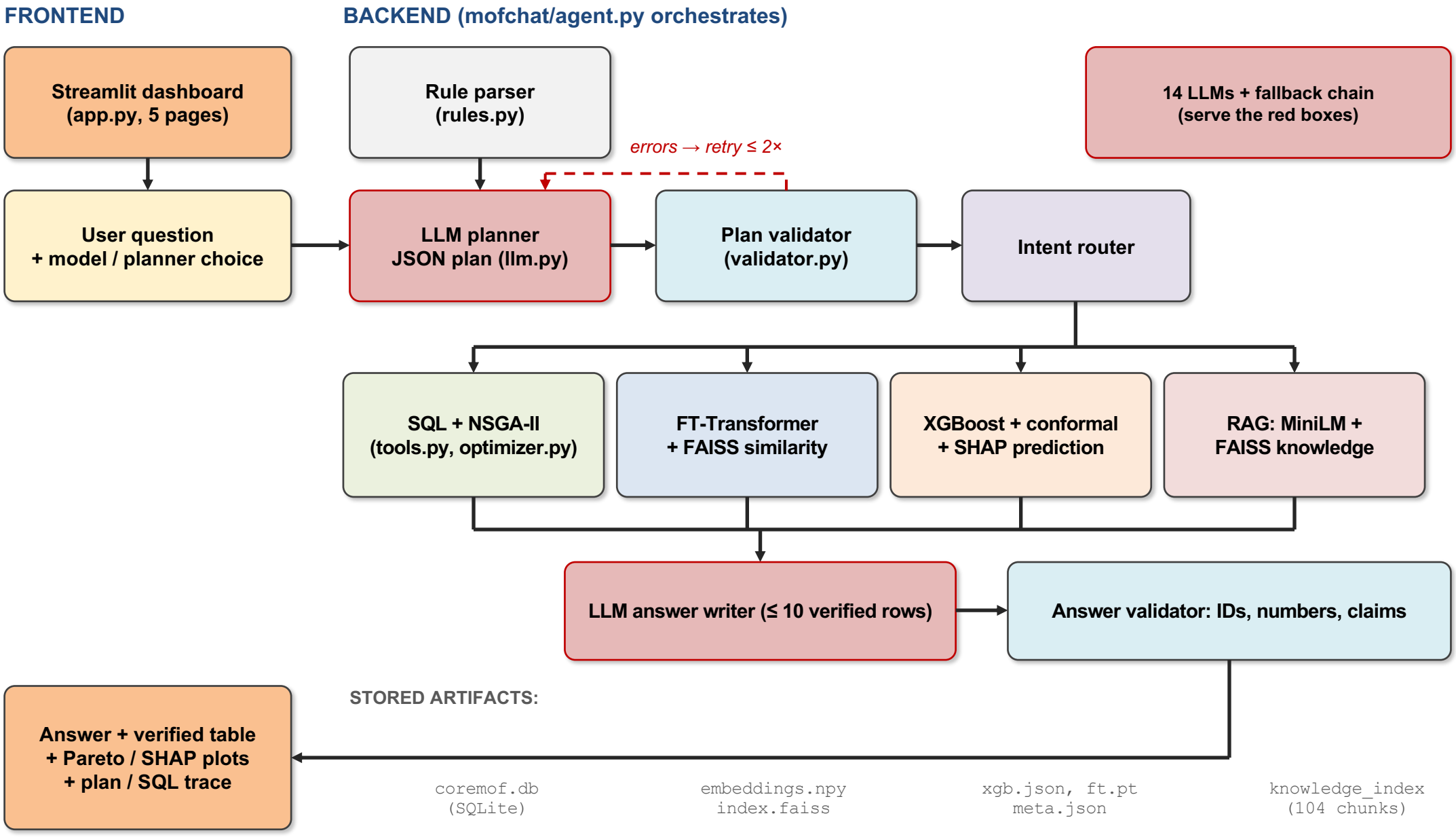






- **PCA:** PC1 + PC2 explain 78 % of the variance of the ten descriptors
- **t-SNE of the 64-d FT-Transformer embeddings:** smooth surface-area gradient; metals form local clusters
- **Masked reconstruction:** surface area R^2 0.978, void fraction 0.913 when hidden from the model
- **SHAP:** pore volume and void fraction drive surface area; density drives void fraction

Overall Pipeline: from Dashboard to Backend



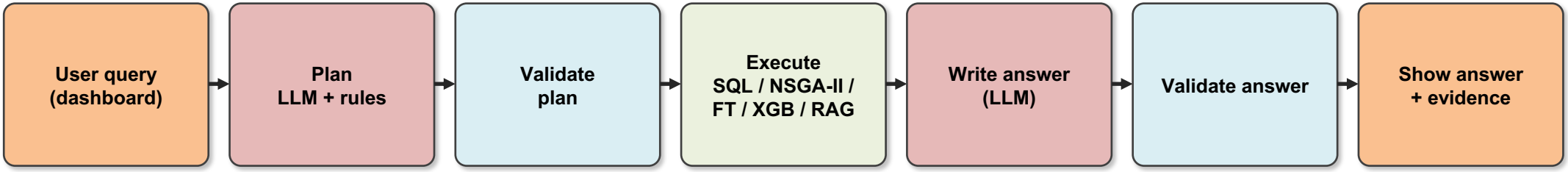
red = uses a language model; other boxes = deterministic, unit-tested code

Pipeline Architecture: Offline Training and Online Inference

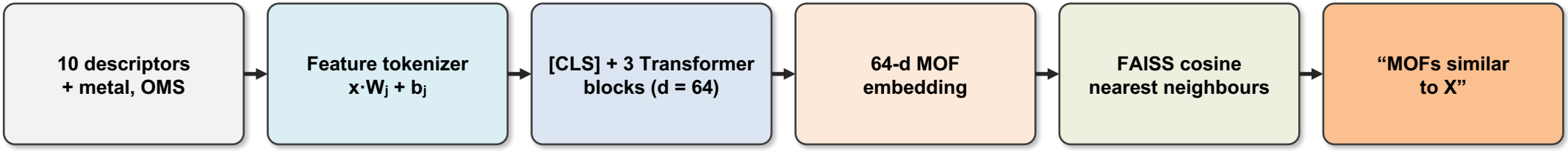
Offline (run once)



Online (per question)



Feature-extraction path inside the FT-Transformer

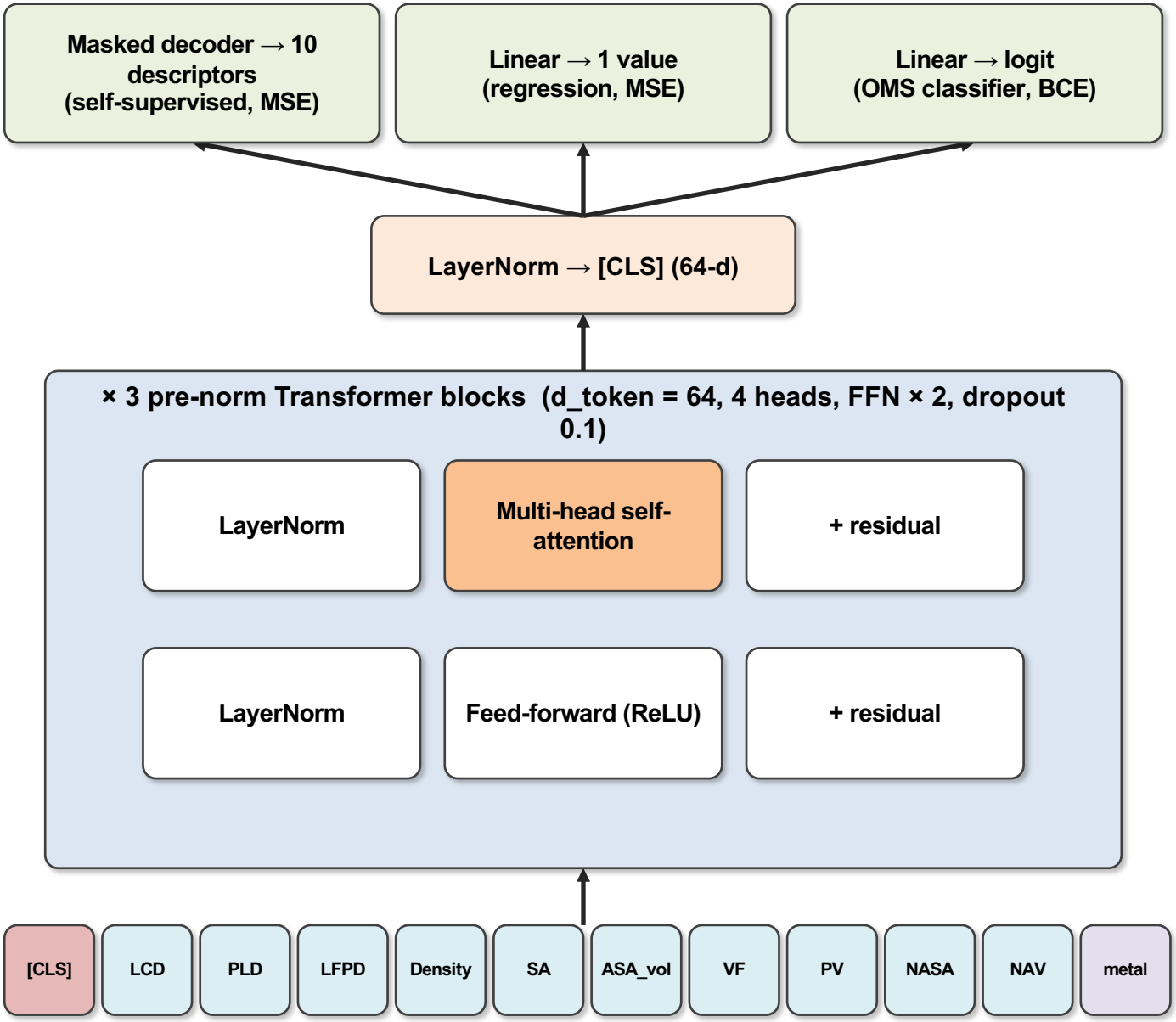


Review-1 used 384-d embeddings matched to a Sentence-BERT question vector; the two were never trained together, so that path was replaced by plan → validate → SQL (9.4 % → 100 % exact answers).

Models and Embedded Models Used

Model / tool	Type	Role in MOFChat
Gemini 3.1 Flash-Lite (default)	LLM, cloud (free)	Planner (JSON plan) and answer writer
GPT-OSS 120B / 20B, Qwen 3.8 27B (Groq)	LLMs, cloud (free)	Fallback planners / writers
Nemotron 3, Gemma 4 (OpenRouter)	LLMs, cloud (free)	Further fallbacks
Llama 3 / 3.1 8B, Qwen 2.5 7B, Nemotron Mini (Ollama)	LLMs, local	Fully offline planning and answers
all-MiniLM-L6-v2 (Sentence-BERT)	Embedding model, 384-d	Embeds questions + knowledge chunks for RAG
FT-Transformer, masked autoencoder (114k params)	Embedding model, 64-d (DL)	MOF embeddings for similarity search
FT-Transformer regressor / classifier	Deep learning	Surface area, void fraction; open-metal-site classification
MLP classifier	Deep learning	Baseline for open-metal-site classification
XGBoost	Gradient-boosted trees	Main surrogate (+ SHAP, conformal); OMS classifier
Gaussian process, Ridge, Logistic regression	Classical ML	Baselines; GP gives 95 % intervals
FAISS	Vector index	Fast cosine nearest-neighbour search
NSGA-II	Multi-objective algorithm	Pareto ranking of candidate MOFs
SHAP (TreeSHAP), split-conformal	Explainability / uncertainty	Feature contributions; 90 % prediction intervals
SQLite, Streamlit, PyTorch, LangChain	Infrastructure	Database, dashboard, DL framework, LLM clients

Model Architecture: FT-Transformer (Gorishniy et al., 2021)



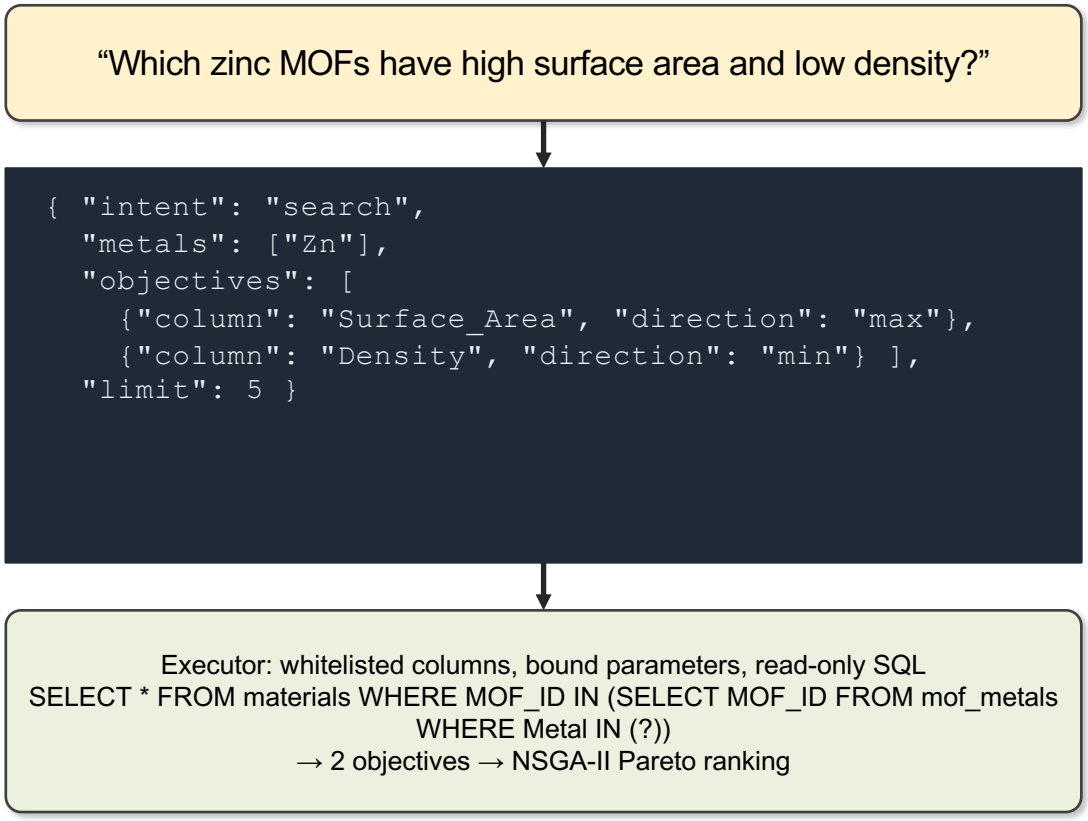
Input tokens: numeric $x_j \cdot W_j + b_j$, categorical embedding, learned [CLS]; 30 % of numeric tokens replaced by [MASK] during self-supervised training

Configuration

Item	Value
Parameters (embedding)	114,378
Parameters (classifier)	104,449
Blocks / heads / d	3 / 4 / 64
Optimiser	AdamW, lr 1e-3, wd 1e-5
Batch / early stopping	256 / val loss
SSL masking	30 % of tokens
SSL training	38 epochs
Regressor (SA)	163 epochs
Classifier (OMS)	best epoch 31

Embedding → FAISS similarity. Regression → surface area / void fraction. Classification → open metal site.

LLM Planning Architecture: Plan, Validate, Execute



Intents classified by the planner: search · count · lookup · similar · predict · knowledge · out_of_scope

Validator checks

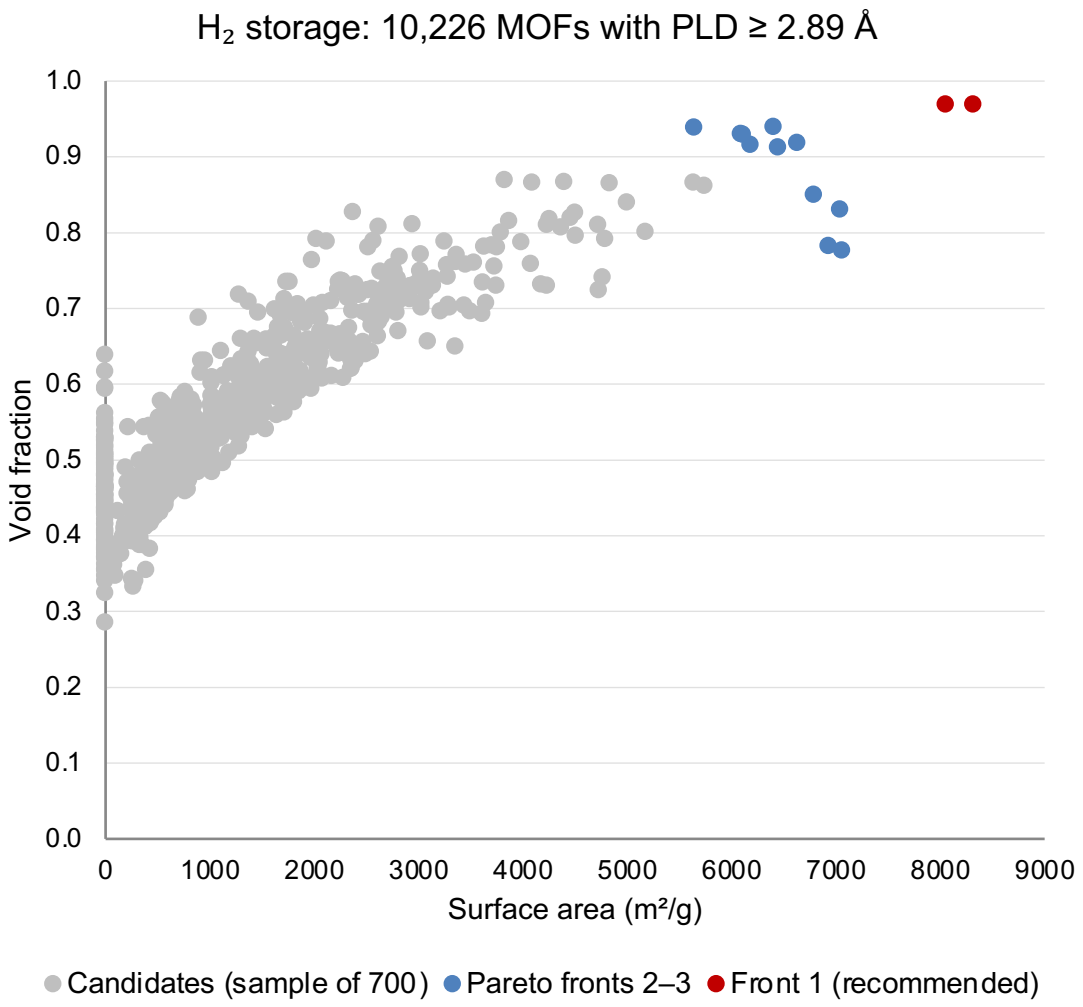
- Column aliases resolved from one registry
- Units, operators and percentages normalised
- Impossible or invented values rejected
- Thresholds not in the question → preferences
- MOF IDs not in the question dropped
- Superlatives written as constraints repaired
- Application presets expanded (H_2, CH_4, CO_2, sieving, catalysis)
- Objectives not asked for in an application question dropped

Fallback: if errors remain after 2 corrections, the rule-parser plan is used; if no LLM is reachable, rules + template answer.

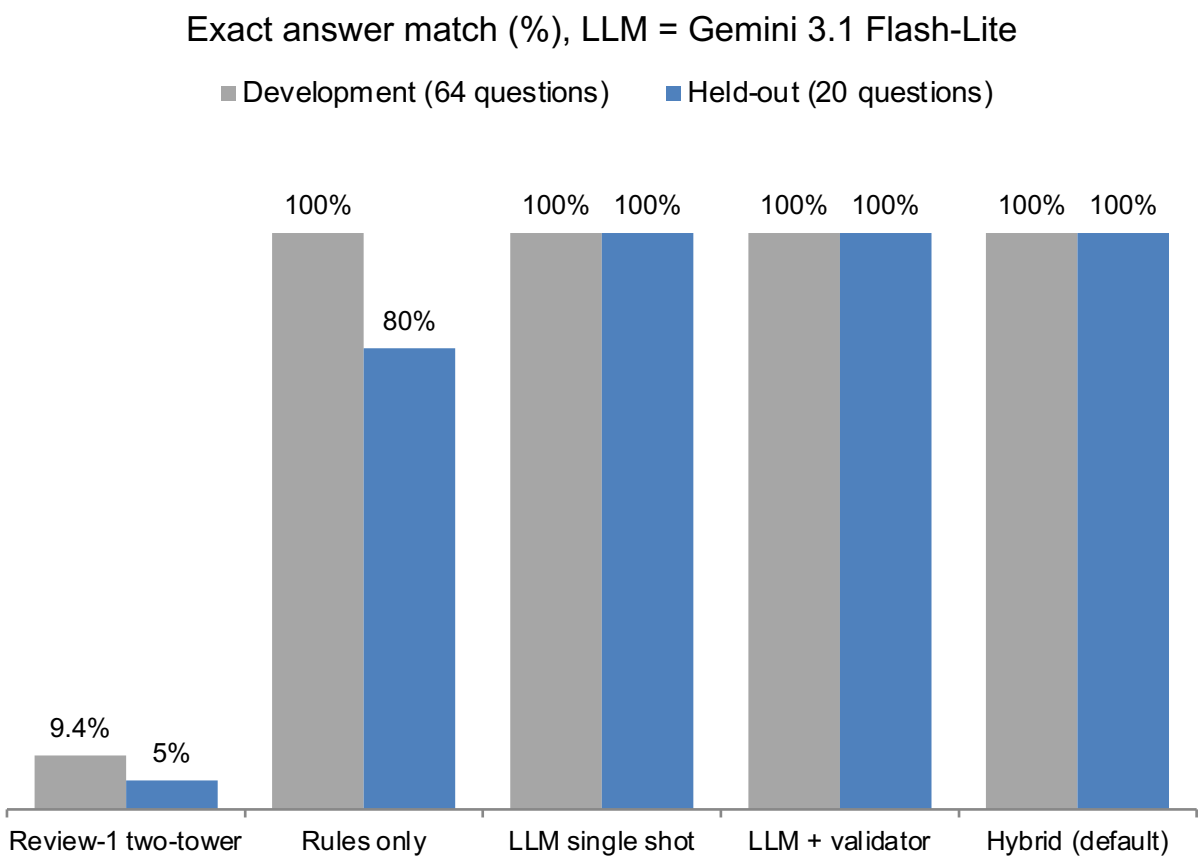
Multi-Objective Ranking with NSGA-II

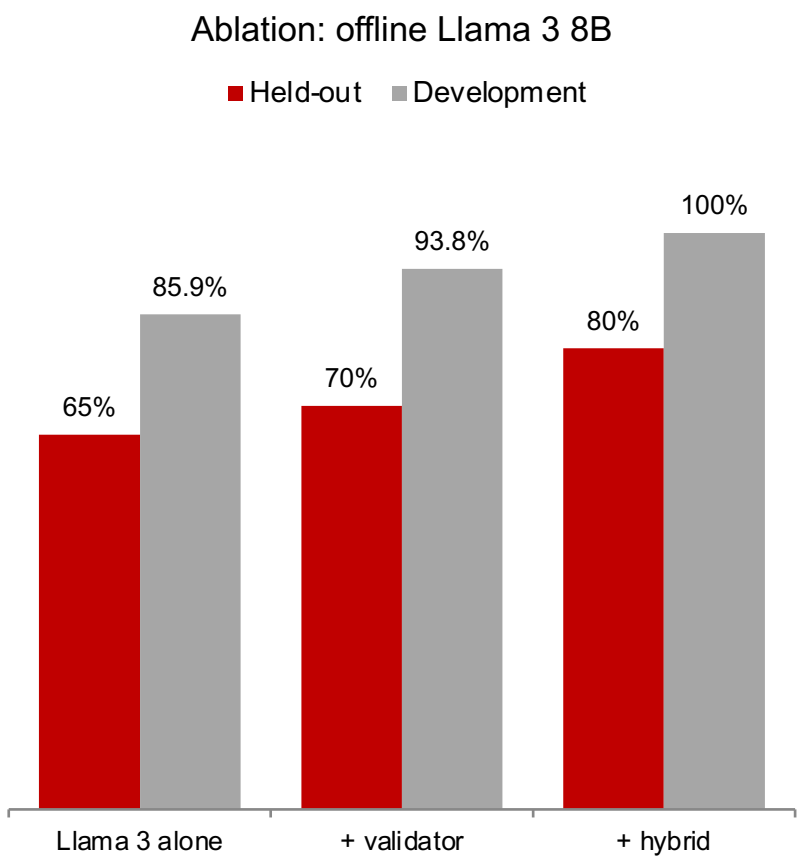
- **Domination:** A beats B if A is at least as good on every objective and better on one
- **Non-dominated sorting:** front 1 = MOFs nobody beats; remove; repeat
- **Crowding distance:** prefers MOFs in sparse regions → diverse trade-offs
- **Exact O(n log n) sweep** for two objectives; worst 25 % screened out first
- **Speed:** all 12,020 MOFs in 0.02 s (2 objectives), 2.6 s (3)

Application preset	Rule (heuristic)
H_2 storage	PLD ≥ 2.89 Å; max surface area, void fraction
CH_4 storage	PLD ≥ 3.80 Å; max volumetric area, void fraction
CO_2 capture	open metal site; PLD ≥ 3.30 Å; min LCD
Separation A/B	d_A ≤ PLD ≤ d_B (kinetic diameters)
Catalysis	open metal site; PLD ≥ 5 Å



Results (1): Retrieval Benchmark — Exact Answers, 5 % → 100 %





- **Exact =** same MOFs (or count) as a hand-written gold plan; constraint precision@k on held-out: review-1 41.8 %, hybrid 100 %
- **Dev 100 % is optimistic** for rules (tuned on it); held-out phrasings were never used for tuning
- **Validator + hybrid add +15 points offline** for < 0.01 s; with a strong cloud model they change nothing (insurance)
- **ChatMOF** reports 96.9 % on its own search set (different questions, not directly comparable)

Results (2): Free-LLM Comparison and Answer Validation

Model	Held-out alone	Held-out hybrid	Dev hybrid	Median call	Provider errors
Gemini 3.1 Flash-Lite (default)	100 %	100 %	100 %	1.5 s	13 / 84
GPT-OSS 120B (Groq)	95 %	95 %	100 %	2.4–4.7 s	0
Qwen 3.8 27B (Groq)	95 %	95 %	100 %	20 s	0
Nemotron 3 Super 120B	95 %	95 %	–	2.0 s	0
Llama 3 8B (local)	65 %	80 %	100 %	1.5 s	0
Gemini 3.5 Flash	–	–	–	–	64 / 84 (quota)

Answer validation (16 full answers per model)

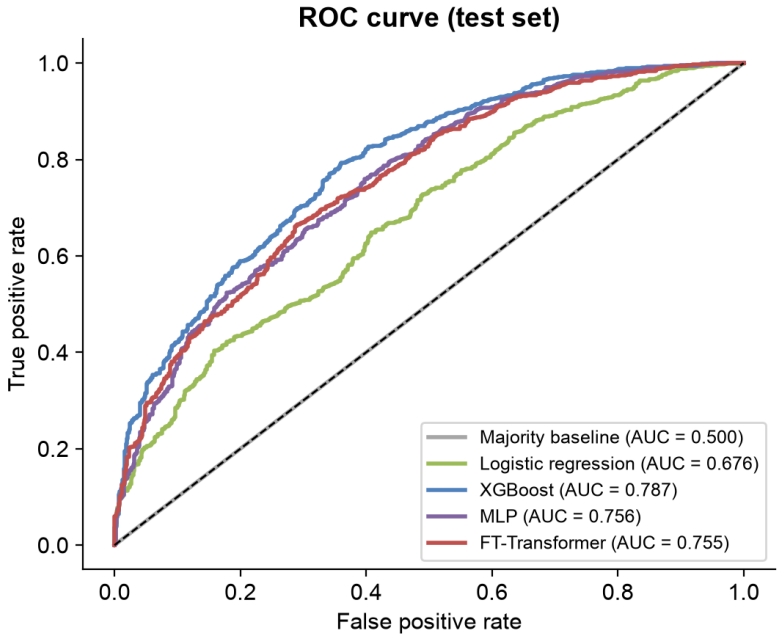
Model	Drafts with a problem	Rewritten	Still flagged	s / answer
Gemini 3.1 Flash-Lite	0 of 16	0	0	9.9
Llama 3 8B (local)	2 of 16	1	0	9.3
GPT-OSS 120B	2 of 16	2	1	30.2
Qwen 3.8 27B	0 of 16	0	0	26.4

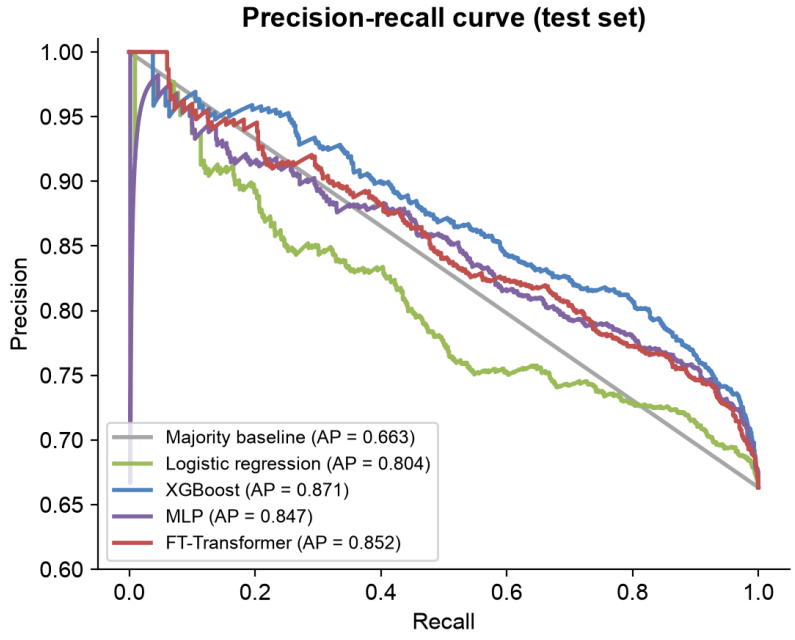
What the validator caught

- Llama 3: invented two densities and “90 %”; ID OFEUN06 → OFERUN06 auto-corrected
- GPT-OSS 120B: claimed “Pareto front 1”; NSGA-II rank was 2 → flagged
- Llama 3 (prediction): invented MOF names (“MOF-210”) → now flagged
- GPT-OSS: cited a prompt header “[WHY]” → now stripped

Results (3): Classification — Open-Metal-Site Prediction (DL vs ML)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Majority baseline	0.663	0.663	1.000	0.798	0.500	0.663
Logistic regression	0.693	0.708	0.916	0.798	0.676	0.804
XGBoost	0.750	0.765	0.901	0.827	0.787	0.871
MLP	0.734	0.759	0.878	0.814	0.756	0.847
FT-Transformer	0.728	0.755	0.873	0.810	0.755	0.852

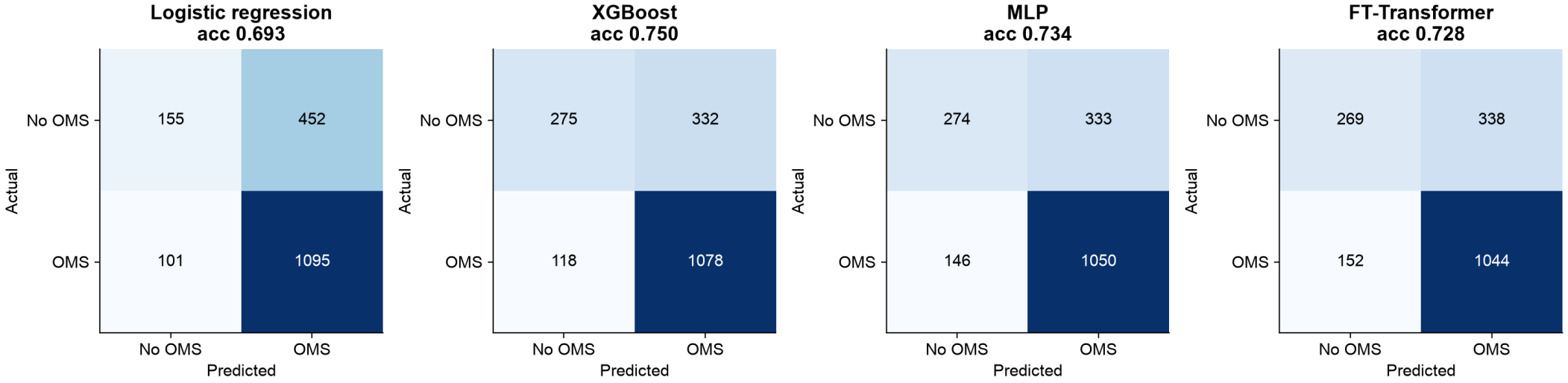




Why this task: an open metal site is required by the CO_2-capture preset; predicting it from geometry lets MOFChat screen descriptor sets that are not in the database.

Split: 8414 / 1803 / 1803 MOFs (train / val / test)

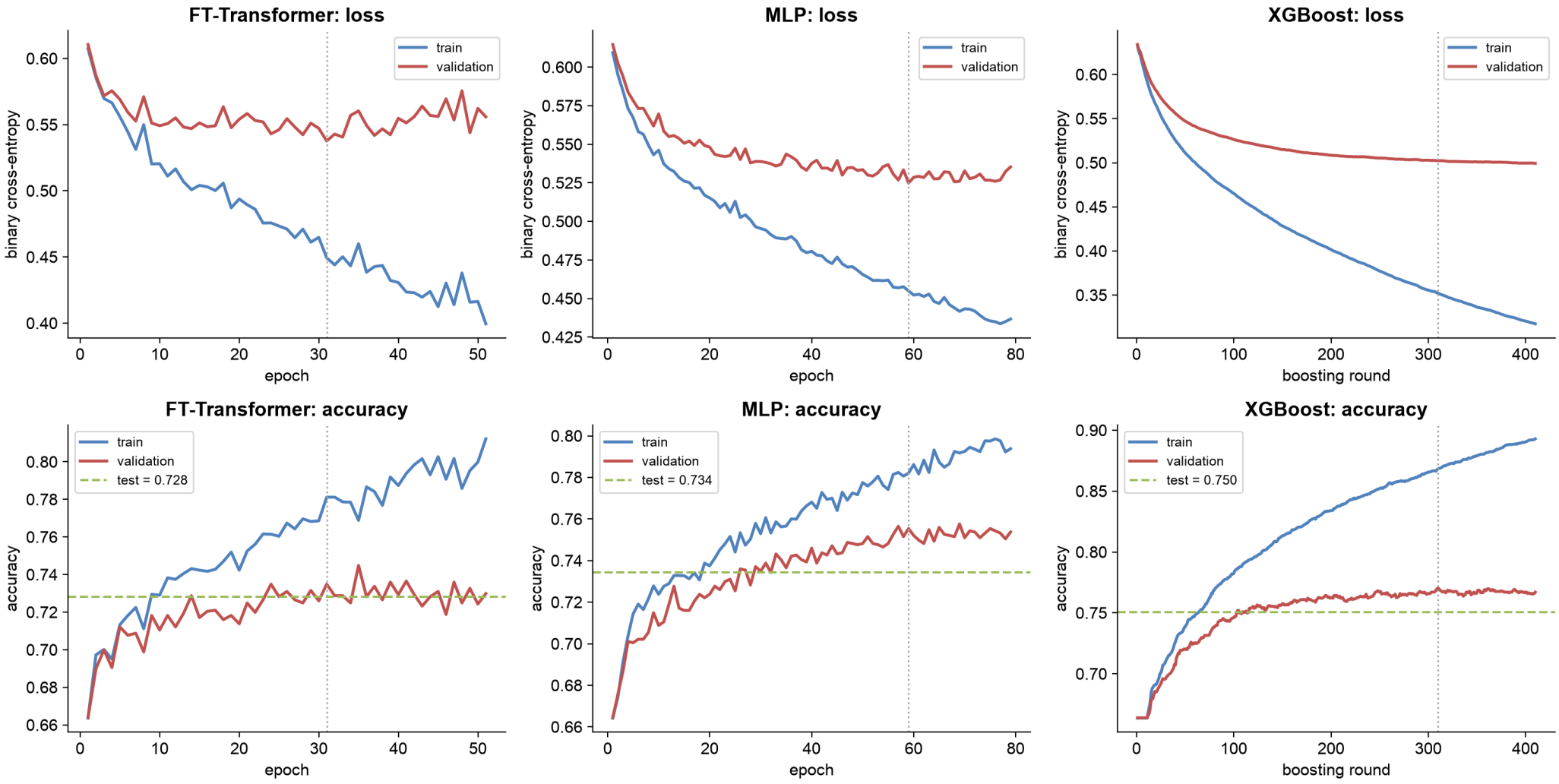
Results (4): Confusion Matrices of the OMS Classifiers (Test Set)



Model	True negative	False positive	False negative	True positive	Train / val / test accuracy
XGBoost	275	332	118	1078	0.868 / 0.770 / 0.750
MLP	274	333	146	1050	0.782 / 0.755 / 0.734
FT-Transformer	269	338	152	1044	0.781 / 0.735 / 0.728

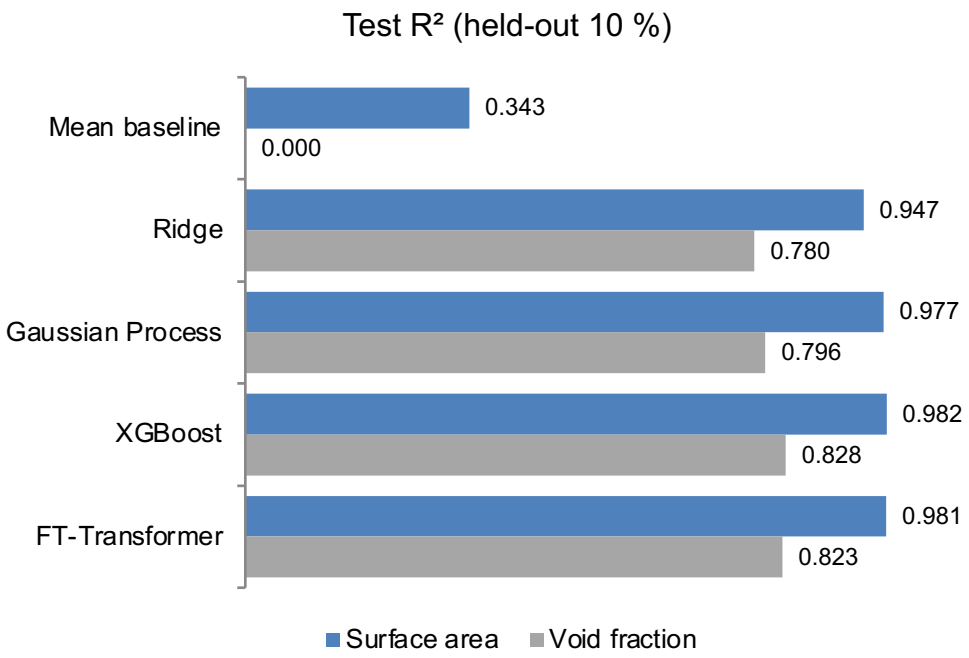
- All learned classifiers beat the majority baseline (66.3 % accuracy, ROC-AUC 0.50); XGBoost is best, the two deep models are close behind
- Train vs test accuracy shows how much each model overfits; early stopping on validation loss controls it for the deep models
- Our reading: whether a metal site is open depends on coordination chemistry that pore geometry and metal type only partly capture, so accuracy levels off near 75 %

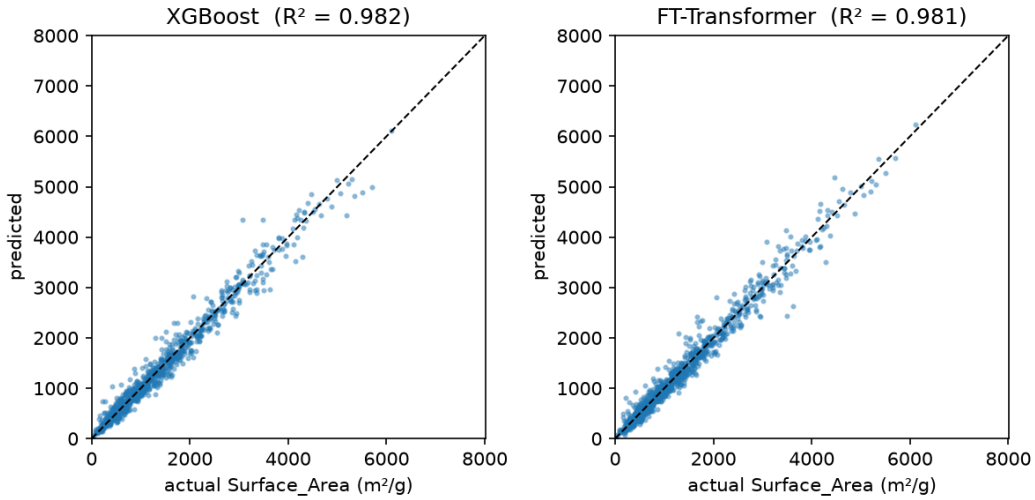
Results (5): Train / Validation / Test Accuracy and Loss Curves



FT-Transformer: best epoch 31 of 51 · MLP: best epoch 59 · XGBoost: best round 310. Validation loss rising after the dotted line = overfitting, which early stopping avoids.

Results (6): Property Prediction (Regression)





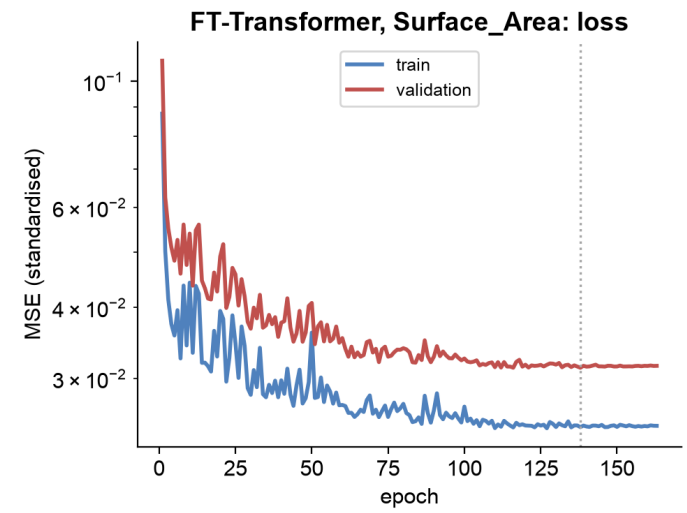
Predicted vs actual surface area on test MOFs

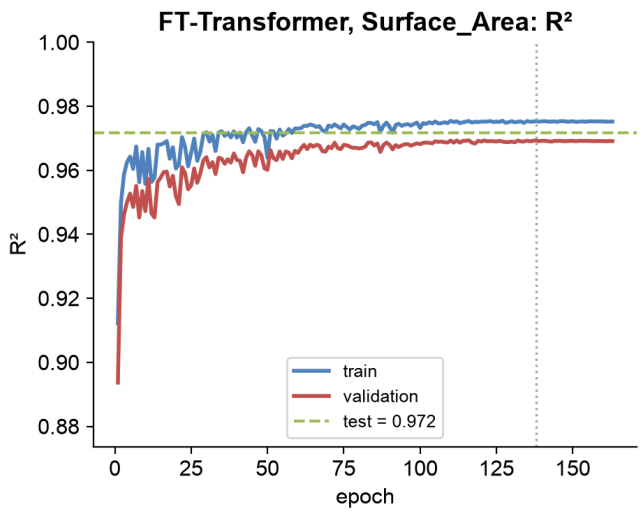
Target	XGBoost MAE	XGBoost R² train / val / test*	FT R² train / val / test*	90 % conformal coverage	GP 95 % coverage
Surface area (m²/g)	90.0	0.987 / 0.959 / 0.972	0.975 / 0.969 / 0.972	90.5 %	93.9 %
Void fraction	0.037	0.899 / 0.820 / 0.828	0.825 / 0.811 / 0.823	88.4 %	95.1 %

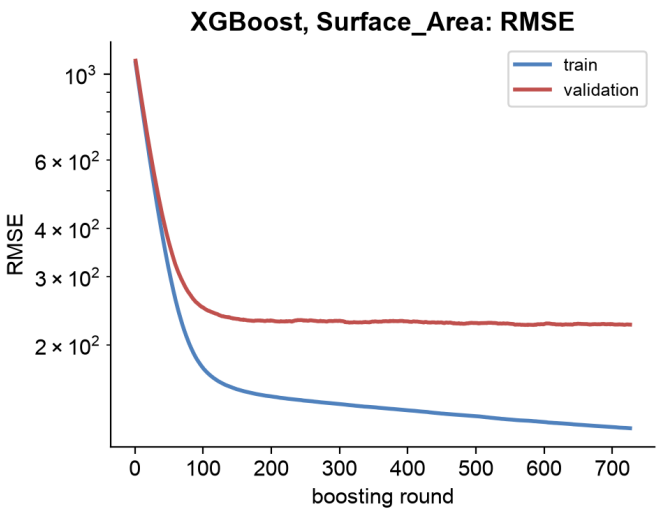
** R² on MOFs with accessible pores (the physics rule handles the zeros exactly; it is 100.0 % accurate on test). XGBoost trains in 0.25 s vs 111.58 s for the FT-Transformer.*

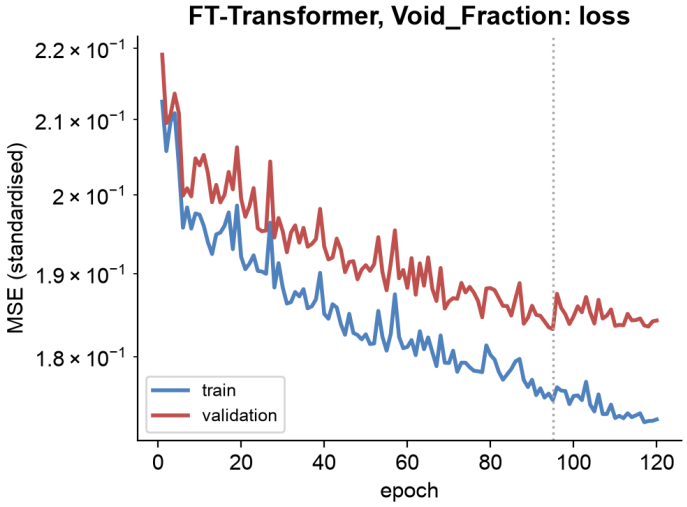
Prediction card in the app: value ± conformal interval, FT-Transformer second opinion, SHAP drivers.

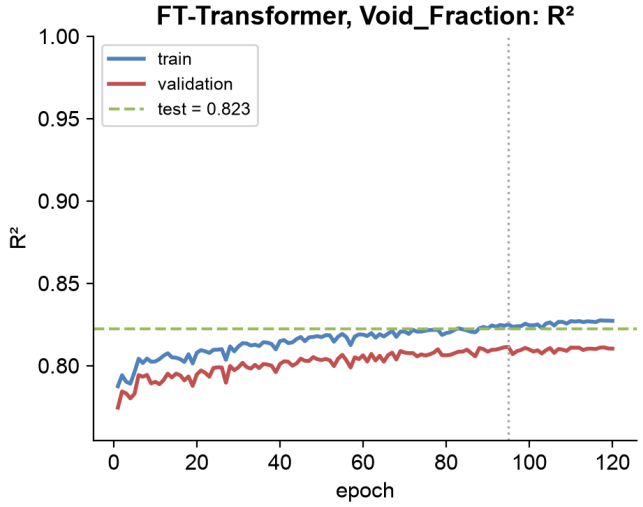
Results (7): Regression Learning Curves — Train vs Validation

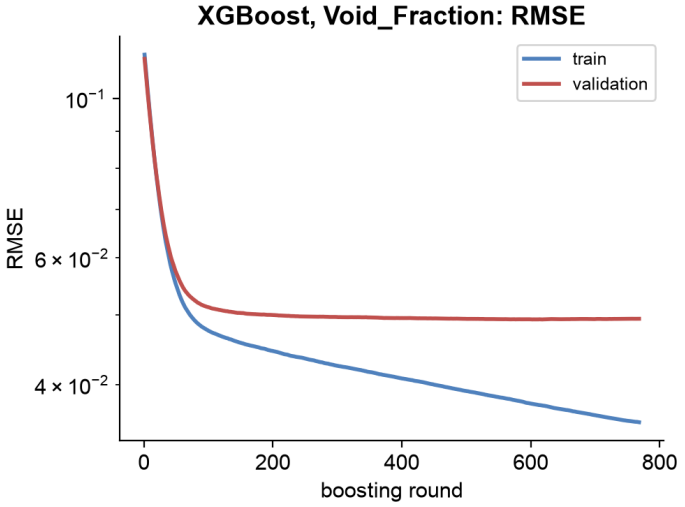






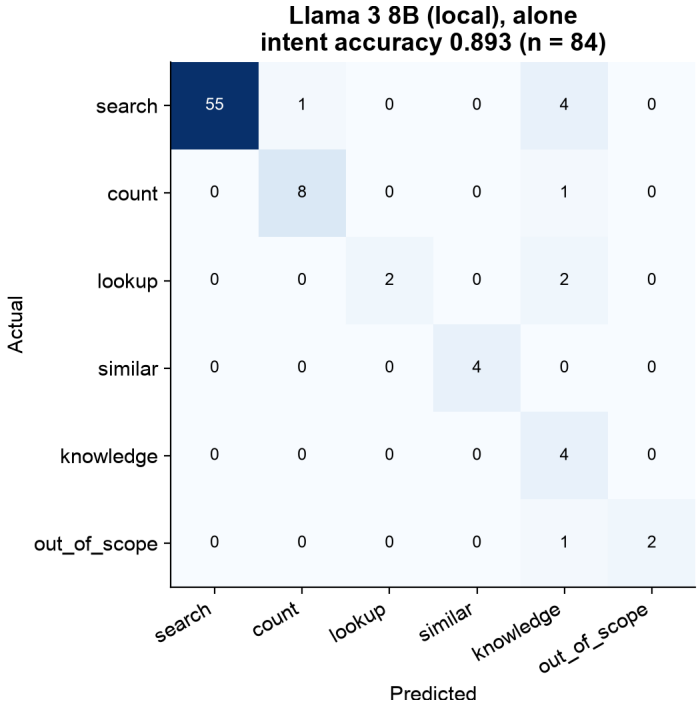


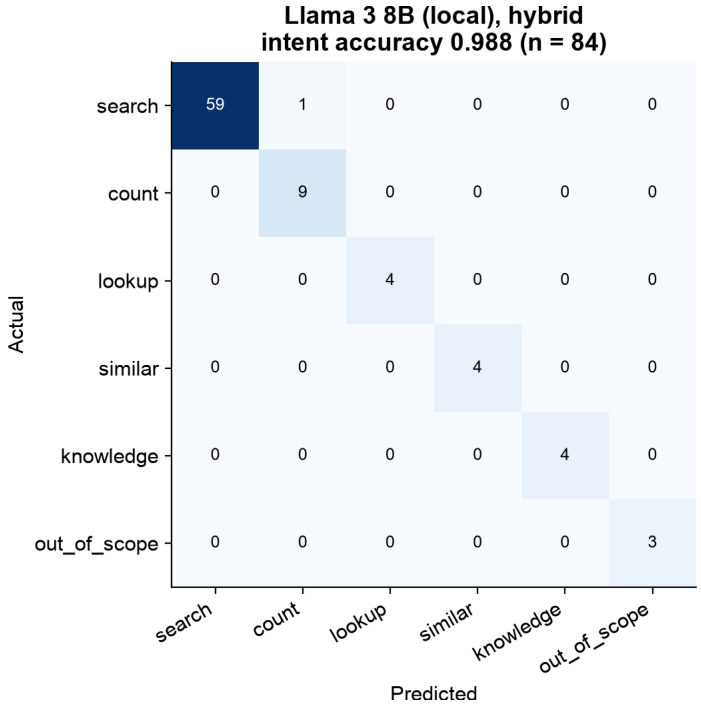


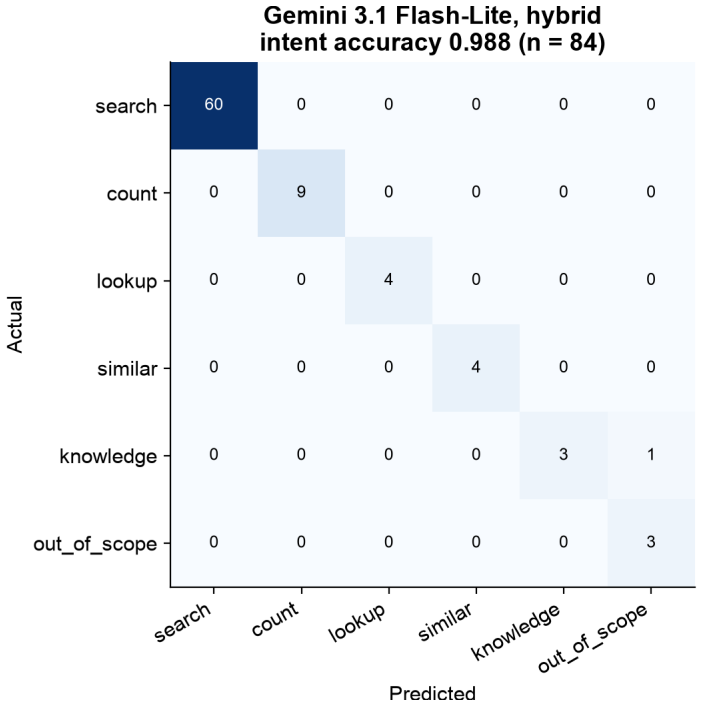


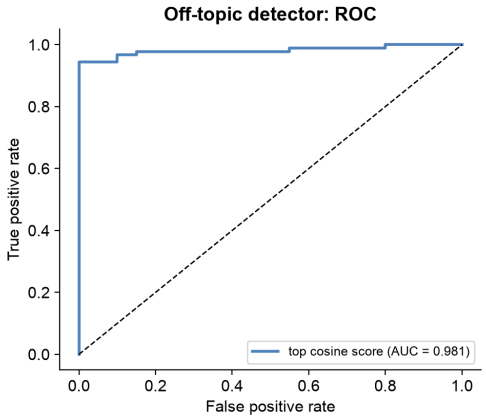
FT-Transformer best epoch: surface area 138, void fraction 95. XGBoost best round: 576 and 618.

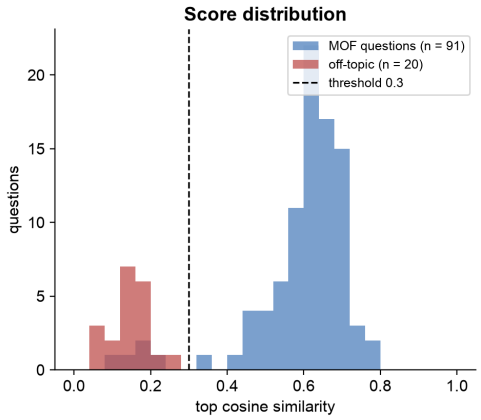
Results (8): Intent Classification and Off-Topic Detection

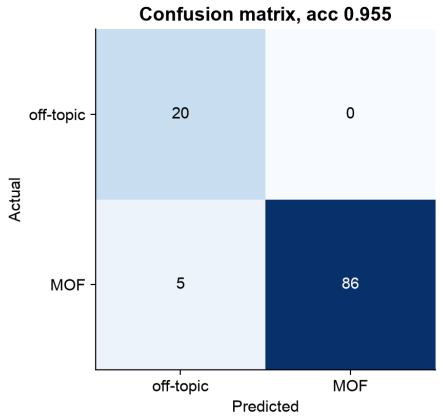








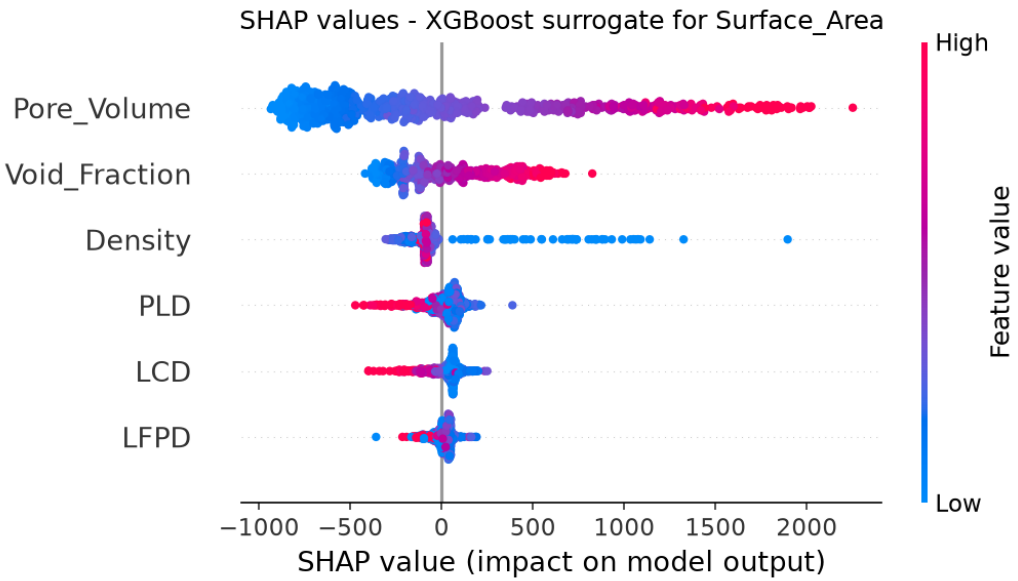


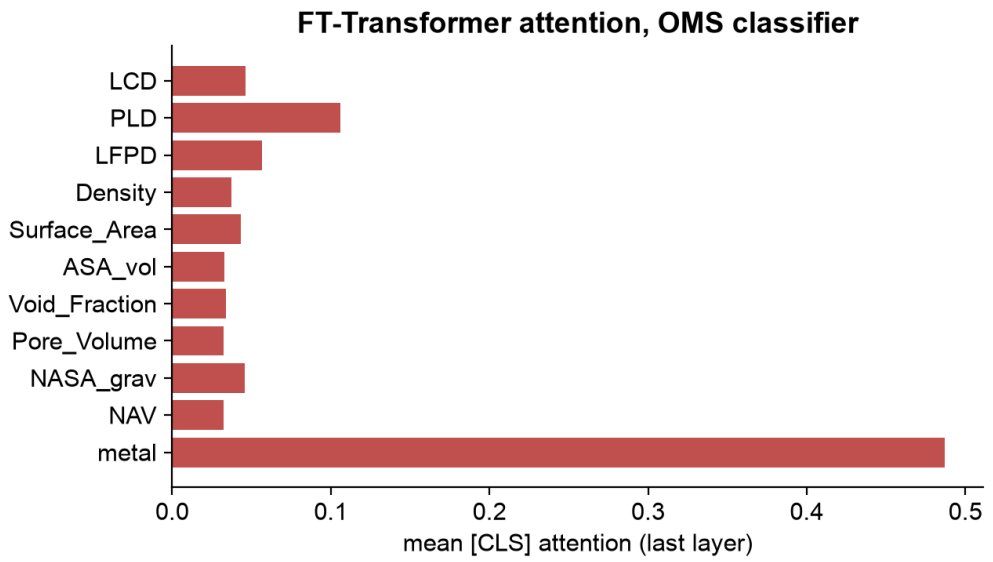


Planner	Intent acc.	Macro F1
Rules planner (no LLM)	0.964	0.950
Llama 3 8B, alone	0.893	0.802
Llama 3 8B, hybrid	0.988	0.990
Flash-Lite, hybrid	0.988	0.952

Off-topic detector: ROC-AUC 0.981, accuracy 0.955 on 91 MOF vs 20 off-topic questions (threshold 0.3, fixed beforehand).

Results (9): Explainability and Similarity Search





Neighbours from (1,000 queries × 10)	Descriptor gap (σ)	Same metal	Same OMS flag
FT-Transformer embedding	0.077	27.8 %	70.2 %
k-NN on raw descriptors	0.068	22.8 %	66.4 %
Random MOFs	0.952	8.0 %	55.7 %

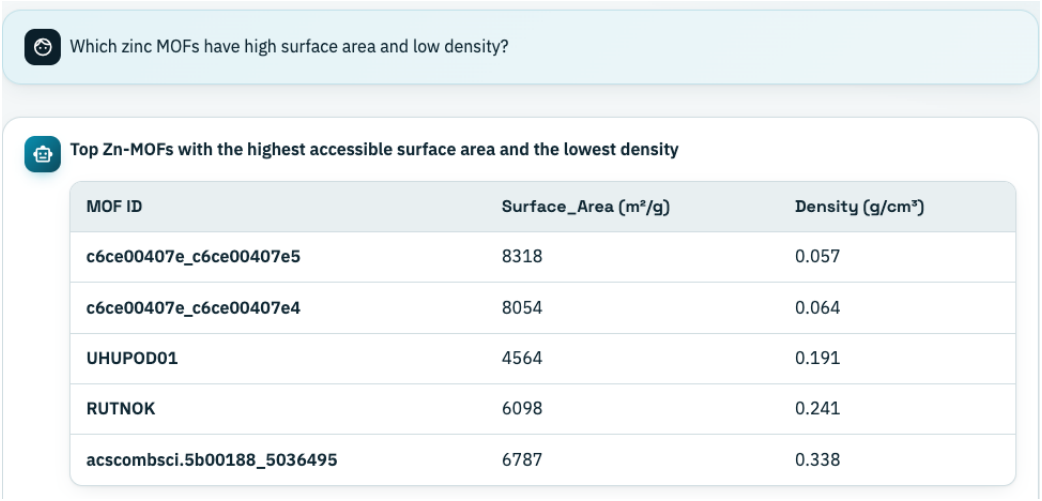
- Pore volume and void fraction dominate surface-area predictions; density dominates void fraction
- Knowledge retrieval: hit@1 75 %, hit@3 83 %
- Every prediction in the app shows its SHAP drivers and a 90 % interval

Results (10): ChatMOF Limitations Addressed

ChatMOF limitation	What MOFChat does	Status
No intrinsic knowledge	RAG over curated notes + ChatMOF paper	**Done**
Token-context limits	SQL first; LLM sees ≤ 10 rows	**Done**
Logic errors in LLM code	JSON plan, validator, whitelisted SQL	**Done**
Restricted search space	All 12,020 MOFs searched and ranked	**Done**
Narrow topology coverage	All CoRE-MOF structures, 65 metals	**Done***
Pretrained model per property	Own surrogates + OMS classifier trained on the table	**Done***
General, non-fine-tuned LLM	Domain prompts, few-shot, presets	**Partly**
High compute cost	Millisecond screening; XGBoost trains in 0.25 s	**Done**
Unvalidated output	Answer validator, intervals, SHAP, benchmark	**Done**

** for recommending existing MOFs and for properties in the table; no structure generation, no gas uptake without uptake labels*

Application: Streamlit Dashboard

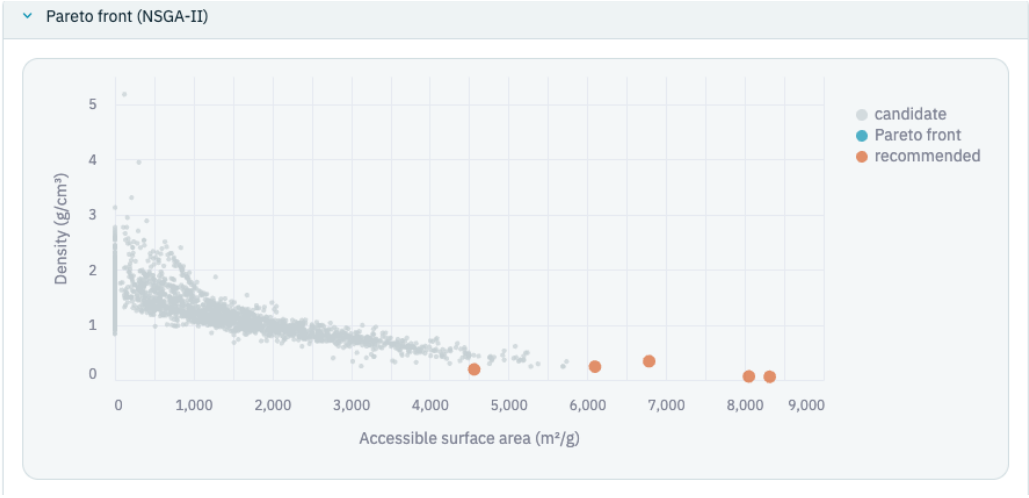


Which zinc MOFs have high surface area and low density?

Top Zn-MOFs with the highest accessible surface area and the lowest density

MOF ID	Surface_Area (m²/g)	Density (g/cm³)
c6ce00407e_c6ce00407e5	8318	0.057
c6ce00407e_c6ce00407e4	8054	0.064
UHUPOD01	4564	0.191
RUTNOK	6098	0.241
acscombsci.5b00188_5036495	6787	0.338

"Which zinc MOFs have high surface area and low density?" — verified table



The same answer's NSGA-II Pareto plot (copper = recommended)

Page	What it shows
Chat	answer, verified table, Pareto plot, SHAP, sources, plan / SQL / validator trace, model used
Explore & optimise	filters + two-objective NSGA-II with Pareto plot
Predict & explain	prediction, 90 % interval, SHAP bars, FT attention
Evaluation	benchmark, model comparison, parity and SHAP plots
About	method summary and data sources

Sidebar: choose any of the 14 language models (or none), planner mode (hybrid / LLM / rules), show plan and validation; system status of every component.

Conclusion and Future Work

Conclusion

- Exact answers over all 12,020 MOFs: held-out 5 % (review 1) → 100 % (default), 80 % fully offline
- Validated LLM planning with 14 free / local models and automatic fallback
- NSGA-II multi-objective ranking with application presets
- Surrogates: surface-area R^2 0.982, 90 % interval covers 90.5 %
- Open-metal-site classification: XGBoost ROC-AUC 0.787, FT-Transformer 0.755, MLP 0.756 (baseline 0.5)
- Answer validator keeps the LLM honest; every number in this deck comes from evaluate.py / evaluate_dl.py

Limitations → future work

Limitation	Next step
No gas-uptake labels	Attach GCMC uptake data; reuse the surrogate pipeline
No structure generation	Diffusion / GA generator (e.g. MOFDiff)
Offline model at 80 %	LoRA fine-tuning of Llama 3 on plan examples
Free-text reasoning unchecked	Semantic-entropy style checks on sentences
Heuristic presets	Learned rules once uptake data exists
Descriptor-only embeddings	Structure / multimodal encoders (L^2M^3OF)

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[3] Y. Gorishniy, I. Rubachev, V. Khrulkov and A. Babenko, “Revisiting deep learning models for tabular data,” NeurIPS, 2021.

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[6] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” NeurIPS, 2017.

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[11] S. Farquhar, J. Kossen, L. Kuhn and Y. Gal, “Detecting hallucinations in large language models using semantic entropy,” Nature, vol. 630, pp. 625–630, 2024.

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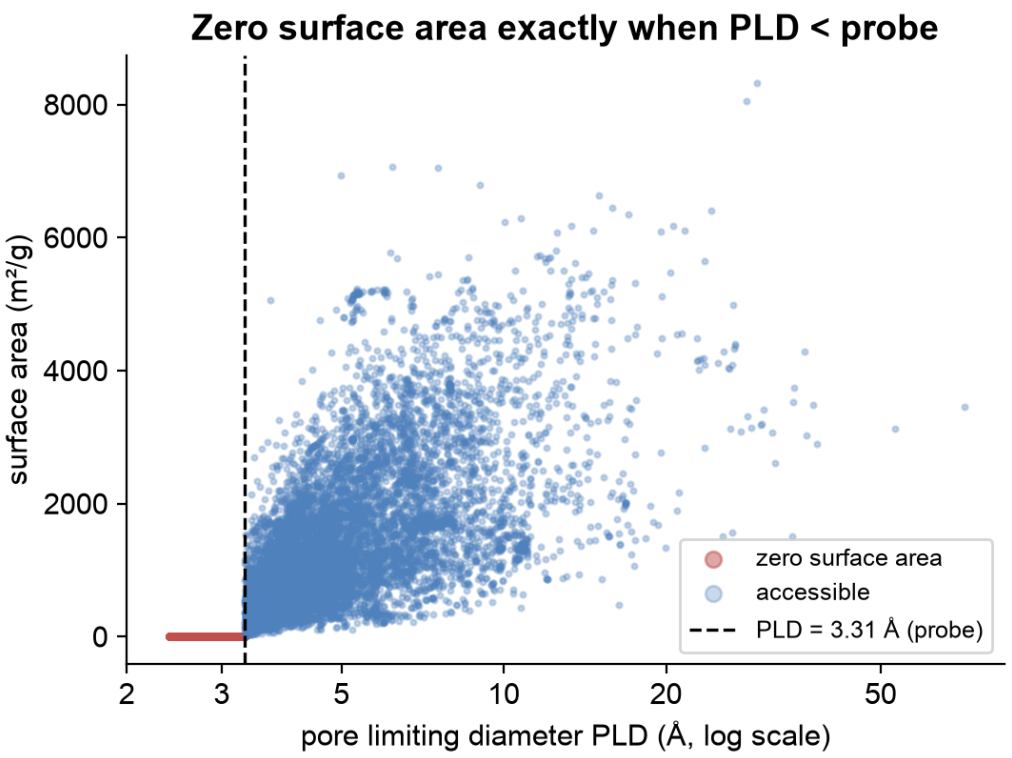
[22] M. Pan et al., “Enhancing spatial reasoning in large language models for MOF structure prediction,” arXiv:2601.09285, 2026.

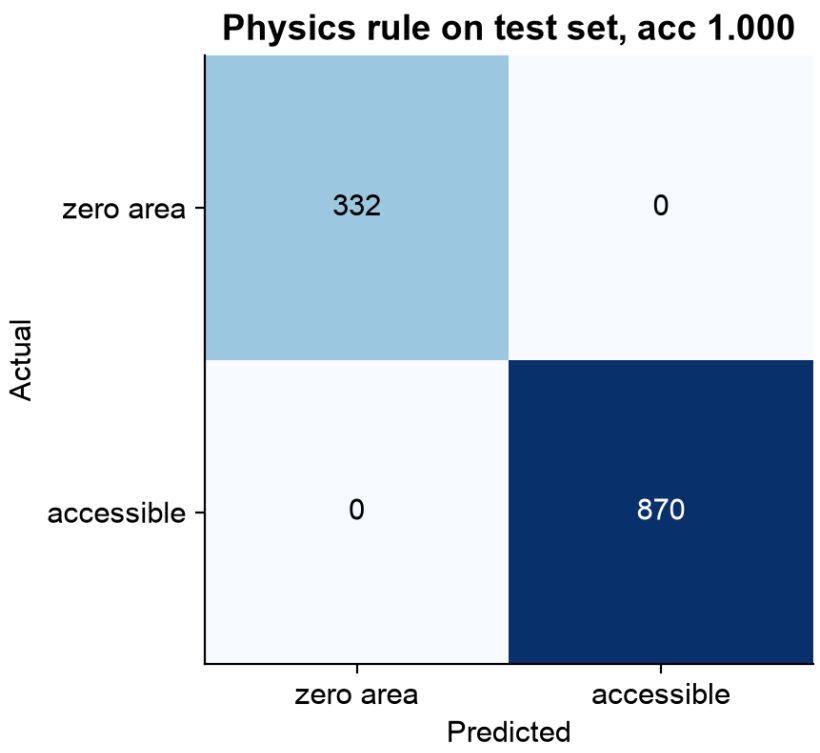
[23] Y. Chen et al., “An LLM agent for end-to-end computational materials discovery,” arXiv:2608.20434, 2026.

Thank You

Questions and live demo · github.com/Sai-Mohith24/MOFChat-AI-Driven-Material-Recommendation-

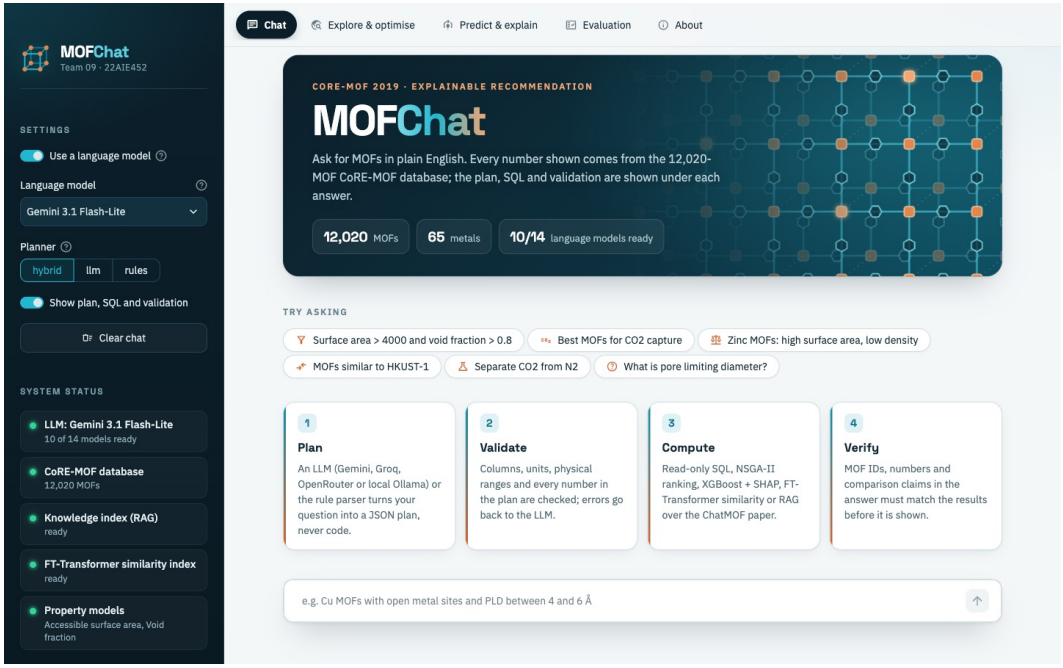
Backup: Physics Rule as a Classifier



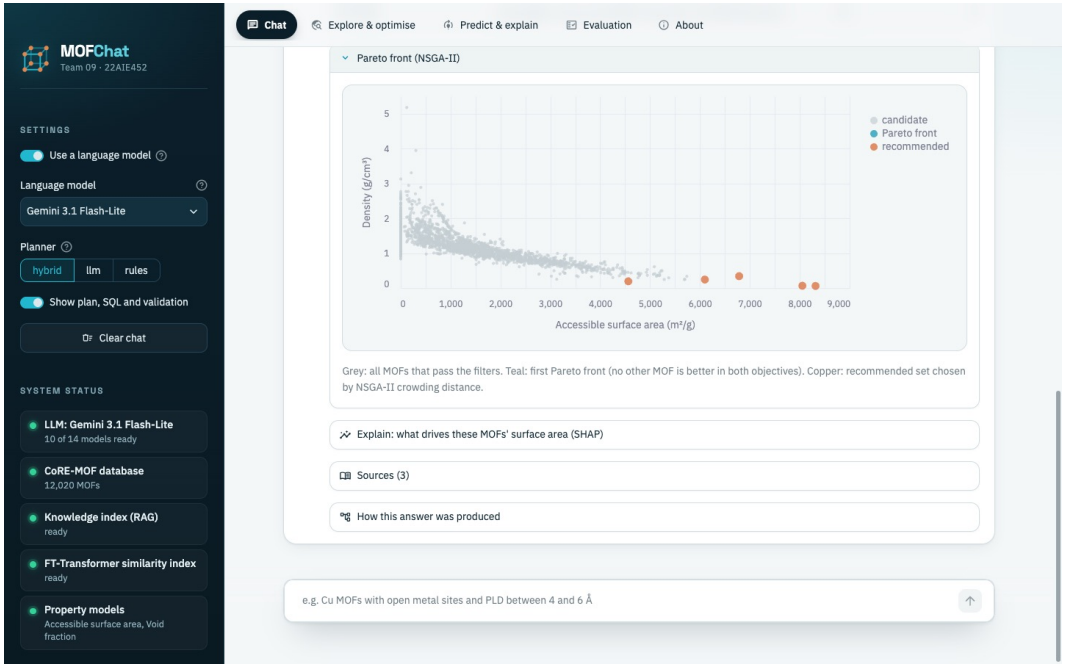


Rule: surface area = 0 if PLD < 3.3107 Å → test accuracy 100.00 % on 1202 MOFs.

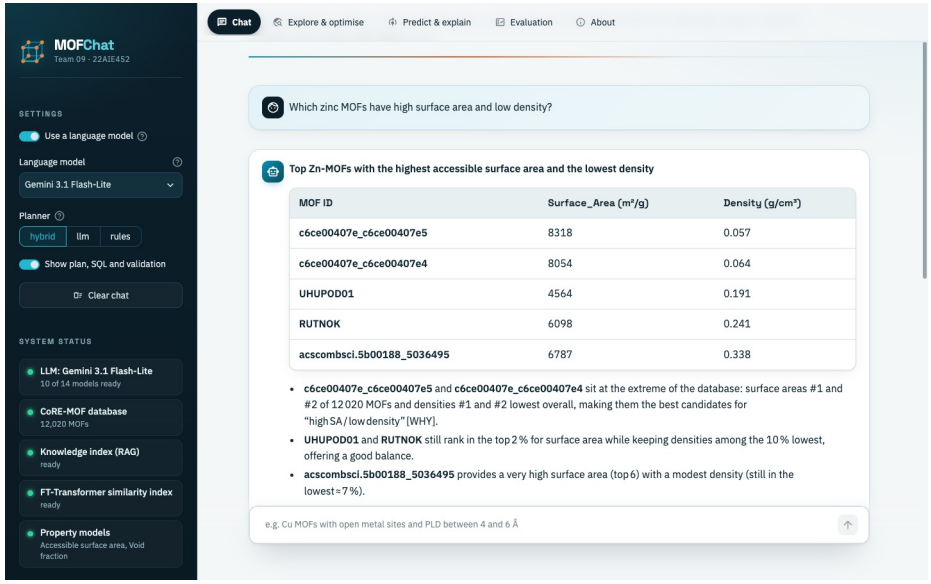
Backup: Demo Screenshots (if the live demo fails)



Chat page: 14 models, planner mode, system status



Pareto plot for the zinc question



Backup: How to Shorten the Talk

- Full talk: slides 1–32 (about 25–30 minutes). 15-minute version: hide slides 2, 7, 8, 14, 17, 20, 25, 26 and 28; keep them for questions
- The 15-minute version still covers: introduction, base paper, objectives, literature, research gap, dataset, feature extraction, pipeline, models used, FT-Transformer architecture, NSGA-II, all main results (benchmark, classification metrics, confusion matrices, learning curves, regression, explainability), application, conclusion, references
- To hide a slide in PowerPoint: right-click the thumbnail → Hide Slide